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Vegetation restoration is affecting the characteristics and patterns of infiltration in the Loess Plateau

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ABSTRACT

In order to improve the fragile ecological environment, the Loess Plateau region of China has implemented a large number of long-term forest vegetation restoration projects. Infiltration, as an important part of the hydrological cycle, is also changing with vegetation restoration, and a comprehensive study of these changes is an essential aspect of evaluating the effectiveness of vegetation restoration. In the study, three typical forest vegetation types in the Loess Plateau were selected to investigate and analyse infiltration characteristics and infiltration patterns by staining tracer experiments. The results showed that (1) vegetation restoration increased the extend and depth of infiltration. (2) The extent of infiltration in mixed plantation forest of *Pinus tabulaeformis* and *Robinia pseudoacacia* was better than that in pure *Pinus tabulaeformis* plantation forest. (3) Vegetation restoration resulted in more pronounced preferential flow phenomenon in the infiltration processes. (4) The preferential flow phenomenon was more pronounced in the natural secondary forest than in plantation forests, especially if the amount of infiltrated water was low. (5) The extent of infiltration was mainly influenced by the infiltration water volume, soil water content, and total soil porosity; and the degree of preferential flow development was mainly influenced by the infiltration water volume, soil organic matter, and root distribution. At present, vegetation restoration on the Loess Plateau can play a role in promoting precipitation infiltration, and mixed-mode afforestation is better than single-dominant-species afforestation to fulfil this function of forests.

1. Introduction

The Loess Plateau is internationally recognized as an extremely fragile ecological environment. Due to the influence of climatic conditions and long-term anthropogenic damage, the local ecological function has been impaired and the environment has deteriorated (Wang et al., 2023). To improve the environment, a large number of long-term vegetation construction projects are currently being implemented in the Loess Plateau (Jia et al., 2020). Under different vegetation restoration patterns, the rainfall redistribution process in forests, such as interception, infiltration, and runoff, will be affected (Chen et al., 2024). Among them, infiltration refers to the process of water entering the soil from the land surface, which is an important part of the hydrological cycle and an important influence factor on soil water storage and vegetation growth (Zhang et al., 2017, Cui et al., 2019). Infiltration characteristics and patterns of different vegetation types are not the same and ultimately lead to different ecosystem hydrological regulation functions, which is a key indicator for assessing the ecological benefits of forests.

Infiltration can be categorized into matrix flow and preferential flow. Matrix flow refers to the relatively slow and uniform movement of water in the soil, which is mainly driven by soil capillarity (Zhao et al., 2022). Preferential flow refers to a more rapid water movement, in which water bypasses the soil matrix region and penetrates rapidly to deeper depths through macropores (Jiang et al., 2017). Due to this rapid infiltration characteristic, preferential flow can contribute up to 15 ~ 85 % to water infiltration, although the preferential flow path accounts for a limited proportion of the total soil pore volume (Zhang et al., 2021). Therefore, many infiltration models have considered this non-equilibrium flow phenomenon to better understand hydrological processes (Zhang et al.,

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Kostiakov model, Philip model, G-A model, etc. are commonly used to model the infiltration process in the Loess Plateau (Cheng et al., 2018, Zhang et al., 2012). Most of these models are based on the full and uniform infiltration characterise of the soil, however, differentiate and mixed patterns of the infiltration was seldom taken into account in these models. Numerous studies have suggested that vegetation restoration greatly increases soil heterogeneity, thereby altering the pattern of uniform water infiltration and allowing a large fraction of precipitation to infiltrate as preferential flow (Fig. 1) (Li et al., 2007). However, it has also been suggested that the surface connectivity of soil macropores is inhibited under vegetation restoration, thus also inhibiting the development of preferential flow (Beven et al., 1982). Therefore, whether or to what extent vegetation restoration changes infiltration patterns may vary in different environments. This requires specific studies of infiltration and preferential flow in the Loess Plateau. The staining tracer method can visualize the water infiltration into the soil, and when combined with image processing technology, it can quantitatively characterize the water infiltration and the development of preferential flow (Liu et al., 2021). Qualitative and quantitative studies of infiltration and preferential flow in the Loess Plateau are the starting point for understanding of water infiltration in local forest soil, as well as of the overall hydrological processes in the forest.

During the development of relevant infiltration models and refinement of research methods, researchers have gained a clearer understanding of the infiltration process and preferential flow phenomena. Due to the heterogeneity and complex interaction of vegetation-soil--water, the mechanism of vegetation restoration and other measures on infiltration characteristics and patterns has been a research hotspot in the field of infiltration (Luo et al., 2019). Numerous studies have concluded that the physicochemical properties of soil affect infiltration, and soil texture, bulk weight, and organic matter are important factors affecting infiltration (Ren et al., 2016, Hao et al., 2019). However, the infiltration described in these studies refers to full infiltration of the soil, whereas recent studies have emphasized the influence of the plant root system on preferential flow, where macropores formed by root channels of living and dead plants can form flow channels for preferential flow (Guo et al., 2019, Wu et al., 2020). In conclusion, numerous experiments and observations have shown that infiltration and preferential flow are affected by a variety of complex factors and vary significantly in different environments (Mei et al., 2018). Therefore, in the Loess Plateau, it is necessary to combine many factors, such as soil physicochemical properties, vegetation characteristics, and hydrological factors, to comprehensively analyse how vegetation restoration affects infiltration characteristics and patterns.

In the study, three typical vegetation restoration types in the Loess Plateau were selected and grassland was used as a control to analyse infiltration and preferential flow phenomena in the Loess Plateau. In order to better understand the effects of vegetation restoration on the characteristics and patterns of infiltration in the Loess Plateau, and to provide a theoretical basis for local forest vegetation management and soil moisture replenishment, the objectives of the study are expected (1) to clarify the general characteristics of infiltration in typical vegetation restoration types, (2) to analyse the development of preferential flow of typical vegetation restoration types, and (3) to elucidate the role of hydrological, soil and vegetation factors on infiltration and preferential flow development.

2. Materials and methods

2.1. Study site

The study was conducted at the Jixian National Ecosystem Research Station of Shanxi Province, China. The station is located in the Caijiachuan Watershed (110°39′45″–169 110°47′45″E, 36°14′24″–36°18′23″N) (Fig. 2). The study area is located in the transition zone between the semi-humid and semi-arid regions in the southeastern part of the Loess Plateau, belonging to the residual loess gully

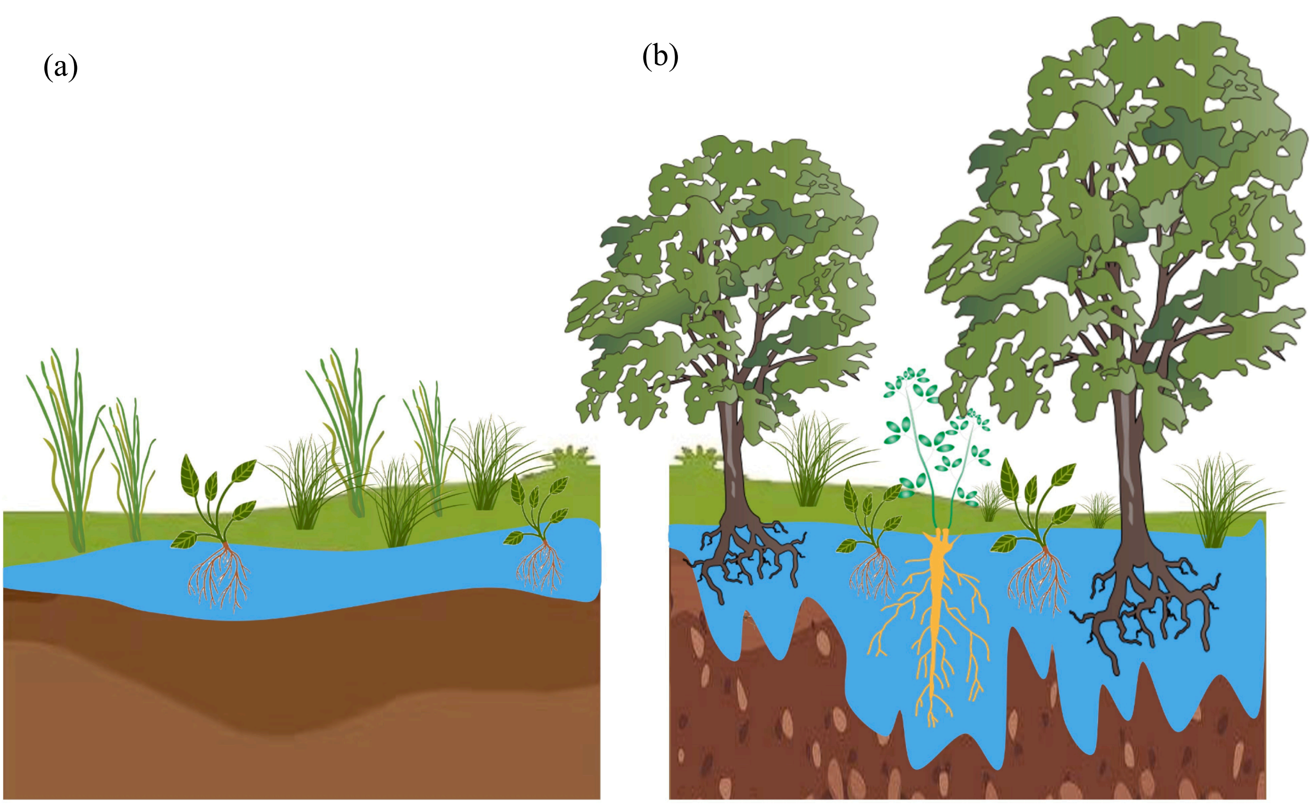

Fig. 1. Effect of vegetation restoration on infiltration. (a) before and (b) after vegetation restoration.

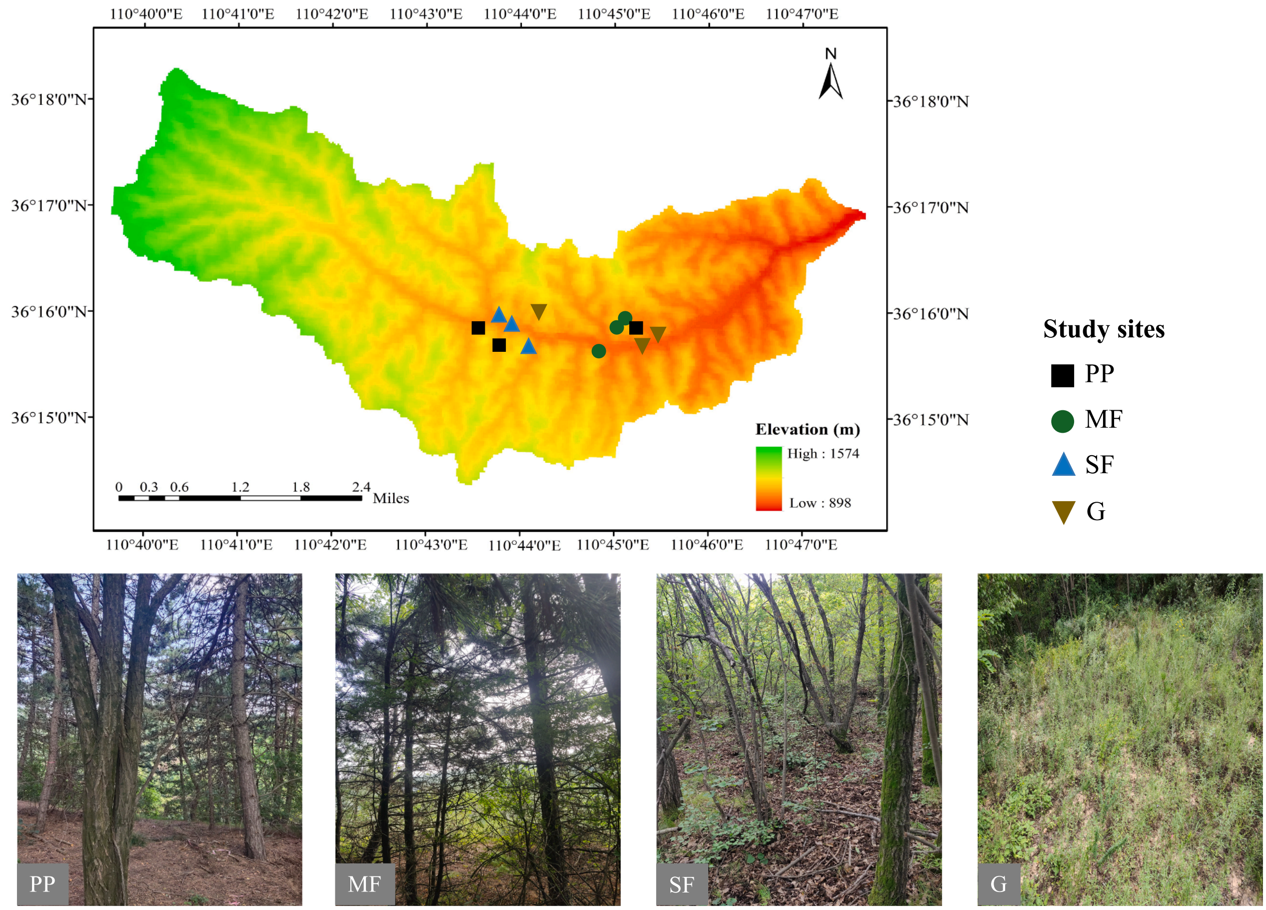

Fig. 2. Study sites location.

area, which is representative of larger scale watersheds on the Loess Plateau. Over the past 70 years, the mean air temperature and precipitation in the study area was 10 °C and 579 mm, respectively. The main soil type in the study area is Haplic Luvisols (soil classification of the Food and Agriculture Organization of the United Nations), which are loess-derived soils and mainly alkaline. The forest cover of the study area is about 75 %, which mainly consists of plantation forests, including *Robinia pseudoacacia* forest and *Pinus tabulaeformis* forest, as well as natural secondary forests.

The study selected typical and representative vegetation restoration types, including *Pinus tabulaeformis* pure plantation (PP), *Pinus tabulaeformis,* and *Robinia pseudoacacia* mixed plantation (MF) and natural secondary forest (SF), and the grassland (G) was used as the control. Three sample plots were set up in flat terrain of each vegetation cover with a size of 10 m × 10 m, and the basic conditions of the topography and vegetation of the sample plots were investigated (Table 1). Four sets of staining tracer experiments were conducted in each sample plot to study infiltration and preferential flow under different rainfall levels (light, moderate, heavy, and torrential). Each set of staining tracer experiments consisted of 5 stained vertical profiles and several horizontal profiles (determined by the actual staining depth).

2.2. *Experiment of infiltration*

The infiltration experiment was conducted with the staining tracer method. The experiment was conducted on the premise that there was no precipitation on that day, the day before and the day after. After determining the location of the staining plots for each group, the ground

Table 1
Basic information of the study sites.

Study Site		Elevation (m)	Forest age (years)	Stand density (plants/hm^2)	crown density (%)	Main vegetation	Dominant understory vegetation
PP	No.1	1147	34	1850	70	*Pinus tabulaeformis*	*Periploca sepium,*
	No.2	1141	34	1550	61		*Sophora davidiia,*
	No.3	1101	34	1575	65		*Rubia cordifolia*, etc.
MF	No.1	1105	34	1300	65	*Pinus tabulaeformis,*	*Rosa xanthines,*
	No.2	1101	34	1650	60	*Robinia pseudoacacia*	*Artemisia sacrorum,*
	No.3	1049	34	1050	52		*Carex humilis Leyss*, etc.
SF	No.1	1092	41	750	30	*Quercus liaotungensis, Populus*	*Rosa xanthines,*
	No.2	1057	41	925	45	*lasiocarpa*	*Artemisia sacrorum,*
	No.3	1050	41	1050	48		*Rubia cordifolia*, etc.
G	No.1	1053					
	No.2	1003					
	No.3	998					

surface was cleared and rectangular metal frames of 130, 60, and 30 cm in length, width, and height, respectively, were smashed vertically into the ground for 5 cm. In-situ ring knife soil samples and bulk soil samples were taken at about 30 cm from each staining plot for measurements of soil physicochemical properties. Samples were collected in layers at 10 cm intervals up to 60 cm of soil (Guan et al., 2023).

In order to significantly differentiate the effects of different infiltration volumes, and taking into account the actual situation of the study area and previous research experience, the four sets of staining tracer experiments in each sample plot were conducted using 5 mm of light rainfall, 15 mm of medium rainfall, 30 mm of heavy rainfall, and 60 mm of torrential rainfall (24-h rainfall in the forest) as the experimental criteria, respectively. Brilliant blue dye with a mass concentration of 4 $g \cdot L^{-1}$ was used as a tracer for the dyeing experiments, and the amount of solution that needed to be configured for each dyeing sample was calculated based on the simulated infiltration water volume and 5 % depletion volume. The brilliant blue solution was uniformly sprayed into the metal frame with a water pump at a speed of 60 mm/h. After spraying, wait for all the water to infiltrate into the soil, ensuring that there was no accumulated water on the surface. Afterwards the metal frame was covered with plastic film to prevent damage from wildlife (Guan et al., 2023).

After 24 h, the plastic film and metal frame were removed, and the stained plots were divided into two parts of 60 cm × 60 cm for excavation of vertical and horizontal profiles. Since moisture transport around the metal frame is more unstable, the stained profile was excavated by removing 5 cm from the boundary, so the actual excavated planar area was 50 cm × 50 cm. The vertical profile was excavated in one layer with a horizontal width of 10 cm, and the horizontal profile was excavated in layers with 5 cm intervals until no stained patches appeared (Fig. 3). The profiles were trimmed after excavation and then images were taken (Guan et al., 2023). During profile excavation, plant root samples were taken with a root auger.

2.3. Infiltration parameter analysis

Stained images were processed using ERDAS IMAGINE 9.2, Photoshop 2021 and Image ProPlus 6.0. After that, the four characteristic parameters of stained profiles, namely, dye coverage, matrix flow depth, preferential flow ratio and length index, were calculated separately (Yao et al., 2017).

(1). Dye coverage.

The dye coverage (DC) is the percentage of the dyed area of the soil profile to the total area of the profile, which can visualize the overall infiltration of the dye solution into the soil. The expression is:

$$DC = \left(\frac{D}{D + ND}\right) \times 100\% \quad (1)$$

where DC is the dye coverage (%), D is the dyed area of the profile (cm^2) and ND is the undyed area of the profile (cm^2).

(2). Matrix flow depth.

Matrix flow depth (U_F) is the depth of the soil corresponding to a dyed area ratio ≥ 80 % in the vertical soil profile. When the dyed area ratio is above 80 %, the infiltration process is mainly matrix infiltration, which can reflect the occurrence of preferential flow.

(3). Preferential flow ratio.

Preferential flow ratio (P_F) is the percentage of the preferential flow region in the total dyed area of the whole vertical profile, that is, the proportion of the dyed area of the region below the matrix flow in the total dyed area. This value can reflect the development degree of preferential flow. The expression is:

$$P_F = \left(1 - \frac{U_F \cdot W}{D}\right) \times 100\% \quad (2)$$

where P_F is the preferential flow ratio (%), U_F is the depth of matrix flow (cm), W is the horizontal width of the soil profile (50 cm in this study) and D is the dyed area of the vertical soil profile (cm^2).

(4). Length index.

Length index (Li) is the absolute sum of the ratio difference of dyed area at the depth of adjacent soil layers after the vertical profile depth is equally divided. The larger the value, the more drastic the spatial variation of the preferential flow. The expression is:

$$Li = \sum_{i=1}^{n} |DCi + 1 - DCi| \quad (3)$$

where Li is the length index, DCi + 1 and DCi are the dye coverage (%) in layer i + 1 and layer i of the soil vertical profile, respectively, and n is the total number of soil layers.

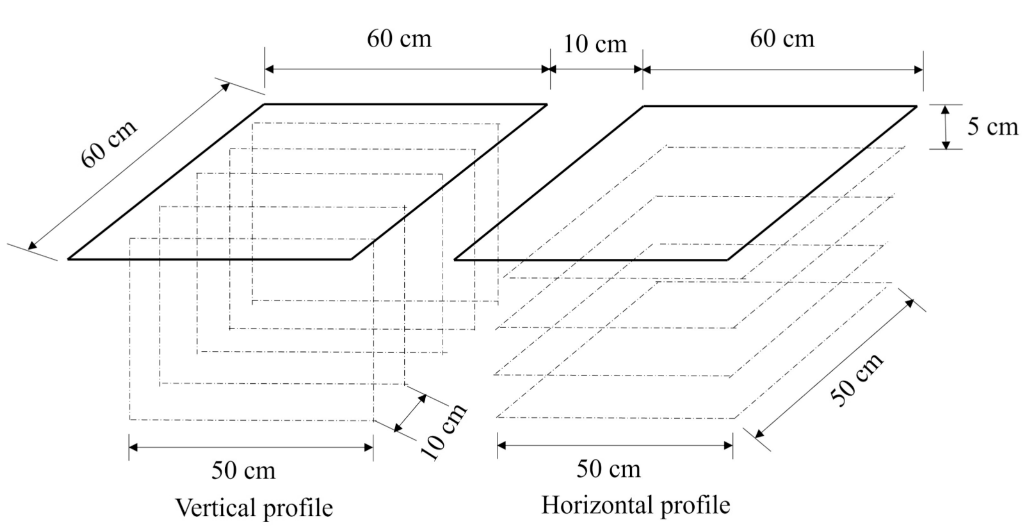

Fig. 3. Schematic of profile excavation.

2.4. Soil and root properties analysis

Soil bulk weight and porosity were determined by the ring knife method. Soil moisture content was determined by the drying method. Organic matter content was determined by the potassium dichromate titration method. Root properties were determined using a digital scanner combined with root analysis software.

2.5. Correlation analysis between infiltration and environmental factors

The infiltration characteristics (DC, U_F, P_F, Li) of each stained plot were matched with the hydrological factors (infiltration volume, initial soil moisture content), soil factors (volume-weight, total porosity, capillary porosity, noncapillary porosity, organic content) and biological factors (root length density, root surface area density, root volume density, root weight density) sampled in that plot, and the relationship between infiltration characteristics and environmental factors was derived by Pearson correlation analysis.

3. Results

3.1. Soil and root characteristics of typical vegetation restoration types

Vegetation restoration affects various properties of understory soils and the distribution of vegetation roots (Fig. 4). Soil moisture content of forest cover was slightly higher than or about equal to that of grassland at the soil layer above 20 cm, and lower than that of grassland at the soil layer below 20 cm. Soil moisture content under the three forests was similar overall, but SF had lower soil moisture content than the two plantation forests at the soil layer above 20 cm.

The forest cover reduced soil bulk density and increased soil porosity. Overall, the total porosity and capillary porosity of natural secondary forest were smaller than that of plantation forests, but its non-capillary porosity was higher. The porosity of understory soil did not monotonously decrease or increase with the soil depth, but the highest values of total porosity and non-capillary porosity of soil in different vegetation cover all appeared in the top 0 ~ 10 cm. The maximum values of capillary porosity were at 20 ~ 30 cm layer of soil in PP and MF, and at 10 ~ 20 cm layer of soil in SF.

Soil organic matter content was generally higher in forests than in the grassland, and organic matter was more aggregated at the surface. Root length density and root surface area density in forests decreased overall with increasing soil layers and was higher in SF than in the two plantation forests. The root surface area density of SF and PP was similar in all soil layers and were significantly higher than those of MF in soil above 20 cm. Root volume density and root weight density of the three typical vegetation types differed significantly from 0 to 20 cm of soil and converged below 20 cm. Overall, SF had relatively greater root volume density and root weight density in the soil above 10 cm.

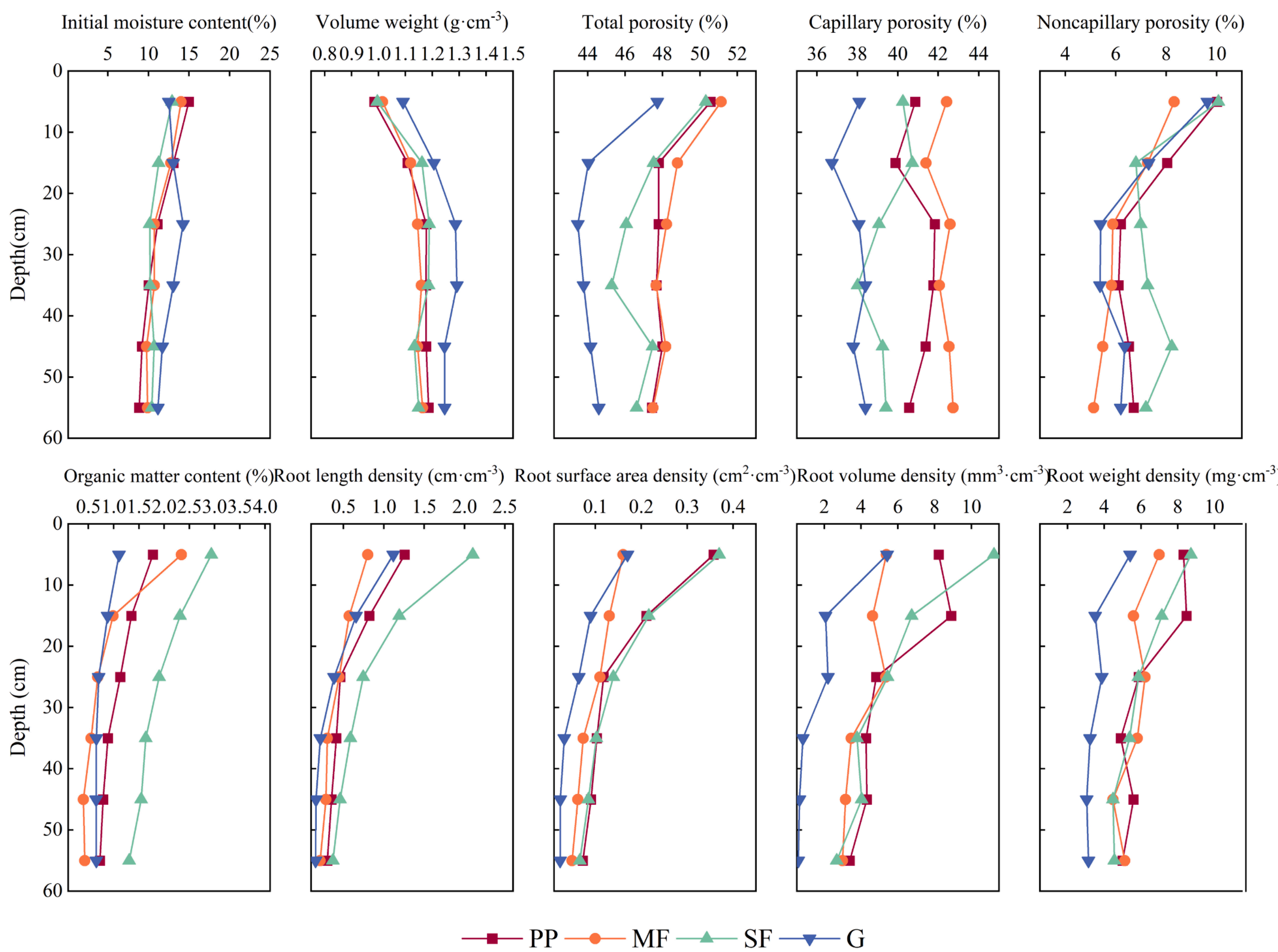

Fig. 4. Soil water, physicochemical and root distribution (error bars are not shown).

3.2. Infiltration characteristics and patterns of typical vegetation restoration types

3.2.1. Extent of infiltration in soil

Plotting the DC curves of vertical soil profiles on a graph provides a visual representation of the overall infiltration of water (Fig. 5). DC of the G was the lowest at any infiltration volume, and DC curve decreased most sharply with soil depth, that is, restoration of forest vegetation increased the range of water movement through the soil, allowing external water sources to recharge deeper layers of soil, which was conducive to infiltration.

At a lower infiltration volume of 5 mm, both PP and MF had DC of 80 % or more at the soil surface (above 5 cm), with water essentially fully infiltrating into the topsoil. As the soil depth increased, DC decreased sharply, and the water absorbed by the deeper soil was greatly reduced. In contrast, SF had a significantly smaller DC at the soil surface than the two plantation forests, but the decrease in DC was slower with increasing soil depth compared to the plantation forests. This implied that in this case water mainly penetrated deeper into the soil in the form of preferential flow along the large pore sizes in soil. However, with the increase of infiltration volume, this phenomenon above was less and less obvious. When the amount of infiltrated water was large, the surface soil under the three typical forest vegetation covers was all in full contact with water. As water continued to infiltrate deeper into the soil, differences in infiltration occurred among the three, as evidenced by the fact that the extent of water infiltration was greatest in soil of MF and smallest in soil of PP.

The DC within the horizontal soil profile varied somewhat with increasing infiltration water volume and deeper soil depth (Fig. 6). Overall, the DC curves of grassland were smoother relative to woodlands, while the DC curves of the three typical woodland soil showed an obvious multi-peaked distribution in most cases. That is to say, there was a clustering phenomenon of water infiltration in woodland soil, and there were different degrees of preferential flow phenomena.

It is worth noting that when the infiltration water was 5 mm and 15 mm, the horizontal soil profile of SF showed an area of DC in the 10 cm soil layer larger than the 5 cm depth, which means that significant preferential transport of water occurred in the soil, and the water had been transported to the deeper soil layer when the shallow layer had not yet formed a homogeneous infiltration. In addition, most of the peaks and troughs of the DC curves in the soil layers at different depths appeared to be synchronized, indicating that the preferential flow was

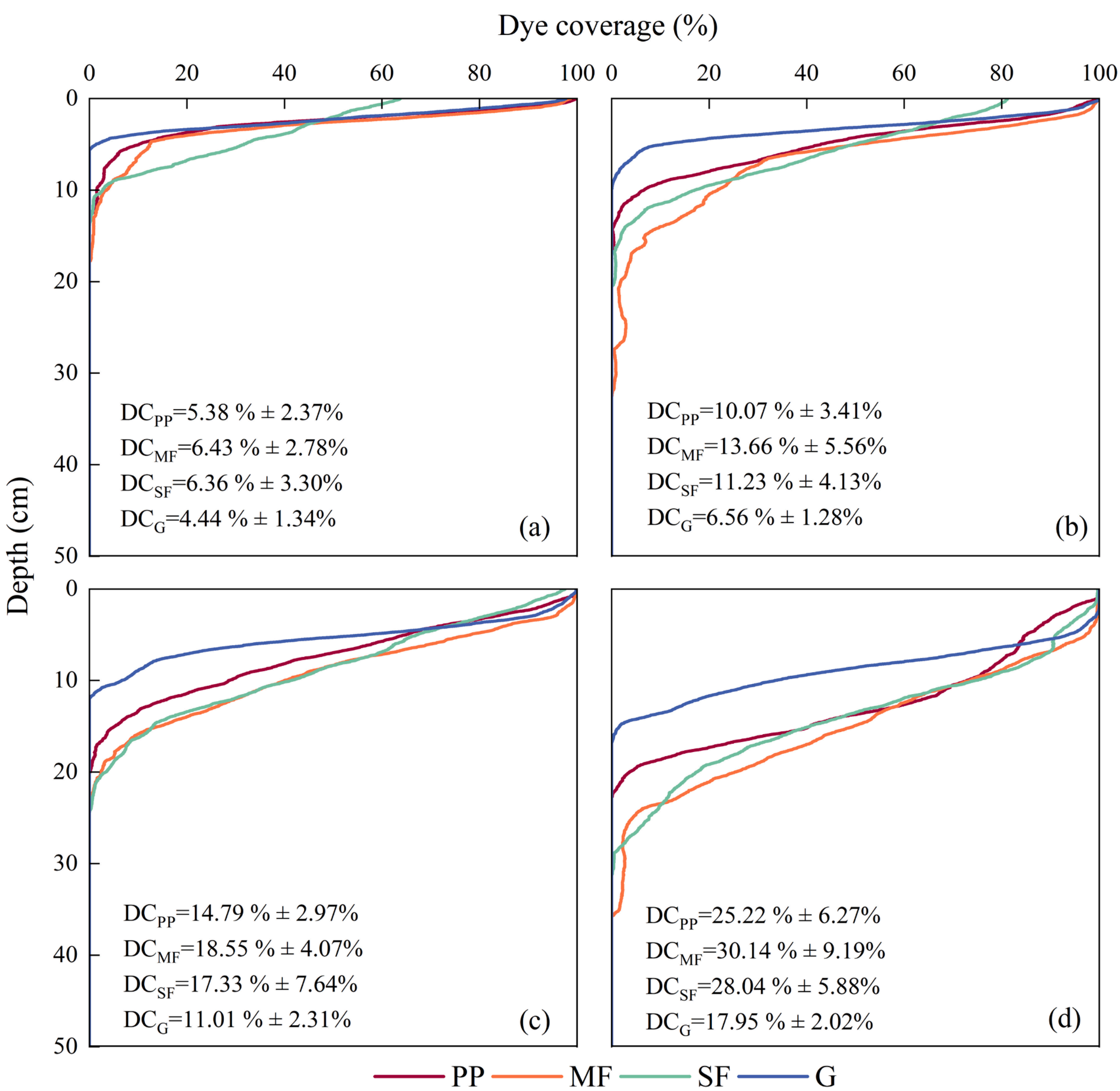

Fig. 5. Extent of water infiltration in soil vertical profile. a, b, c, d represent infiltration volumes of 5 mm, 15 mm, 30 mm and 60 mm, respectively.

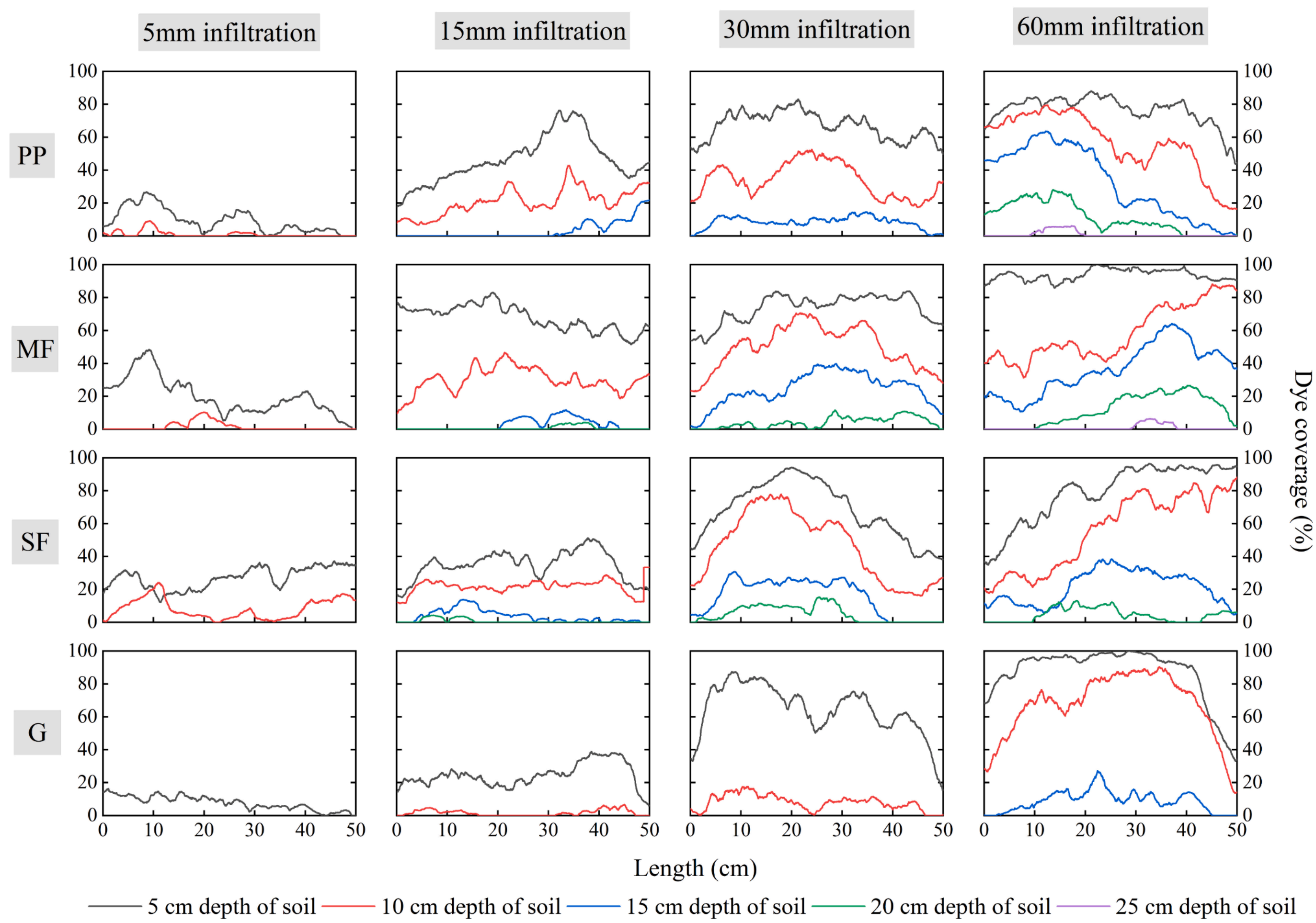

Fig. 6. Extent of water infiltration in soil horizontal profile.

mainly transported to the deeper layers in the vertical direction, with fewer horizontal preferential paths in the soil and less lateral infiltration.

Combining the stained images of vertical profiles provides a more comprehensive understanding of water infiltration under typical vegetation cover (Fig. 7). Because of the large number of stained profiles collected, only one representative stained profile from each sample plot was selected for analysis in this paper.

Water infiltration varied markedly between soil spaces and external conditions, but all infiltration was non-uniform, that is, there was preferential flow. Compared to woodlands, grassland infiltration was much more uniform, and its infiltration area was relatively smaller for the same amount of infiltrated water. Compared with the two plantation forests, the non-uniformity of infiltration in SF was significantly higher, with more obvious preferential pathways in the soil profiles. When the infiltration volume was 5 mm and 15 mm, the staining did not completely cover the surface layer of the soil profile, that is, the infiltrated water did not completely infiltrate the soil matrix in the surface layer, but was directly transported to the deeper soil along the preferential flow paths. When the infiltration volume increased, the surface soil was able to fully absorb the water. When the infiltration volume was 60 mm, although the infiltration was still non-uniform, no obvious preferential flow path could be observed in the soil profile. Water infiltration in plantations was also non-uniform, but the preferential flow was mostly in the form of “funnels” and short clusters, and longer “dendritic” preferential flow did not occur. Water infiltration was more non-uniform in the soil of PP than in the MF, and at 60 mm infiltration, disjointed “patchy” colouring areas appeared in PP soil.

3.2.2. Depth of matrix infiltration, development degree and variability of preferential flow

U_F directly reflects the depth at which water can penetrate uniformly into the soil (Fig. 8). Overall, the depth of uniform water infiltration was deeper in MF, and the soil–water interaction was stronger, allowing for more complete absorption of external water sources. The depth of uniform water infiltration was overall lower in grassland than in woodlands, and the difference was most pronounced at an infiltration volume of 60 mm. U_F in SF was lower than that of plantation forests when the infiltration volume was 5 mm and 10 mm. Even when the infiltration volume was 5 mm, the U_F of SF was 25.7 % lower than that of G. When the infiltration volume increased to 30 mm and 60 mm, the U_F of SF increased and was higher than that of PP, but was always lower than that of MF.

P_F can most intuitively reflect the degree of development of preferential flow (Fig. 8). Overall, PF was higher in SF. The degree of development of preferential flow was higher in SF than in PP and MF. The degree of development of preferential flow was lower in grassland than in woodlands. There was no significant difference in P_F between the two plantation forests at infiltration levels of 5 mm, 15 mm and 30 mm, which were all in the range of 45 %~50 %. It was only when the infiltration volume was 60 mm that the P_F of the plantation forests showed a relatively significant decrease. The PF of SF was as high as 78 % at 5 mm infiltration volume, which was 62.5 % higher than that of PP and MF. As the infiltration volume increased, the P_F of SF decreased, and at 60 mm infiltration volume its P_F was only 5.71 % (PP) and 15.6 % (MF) higher than that of the plantation, which was 0.47 times higher than that of its own at 5 mm infiltration volume.

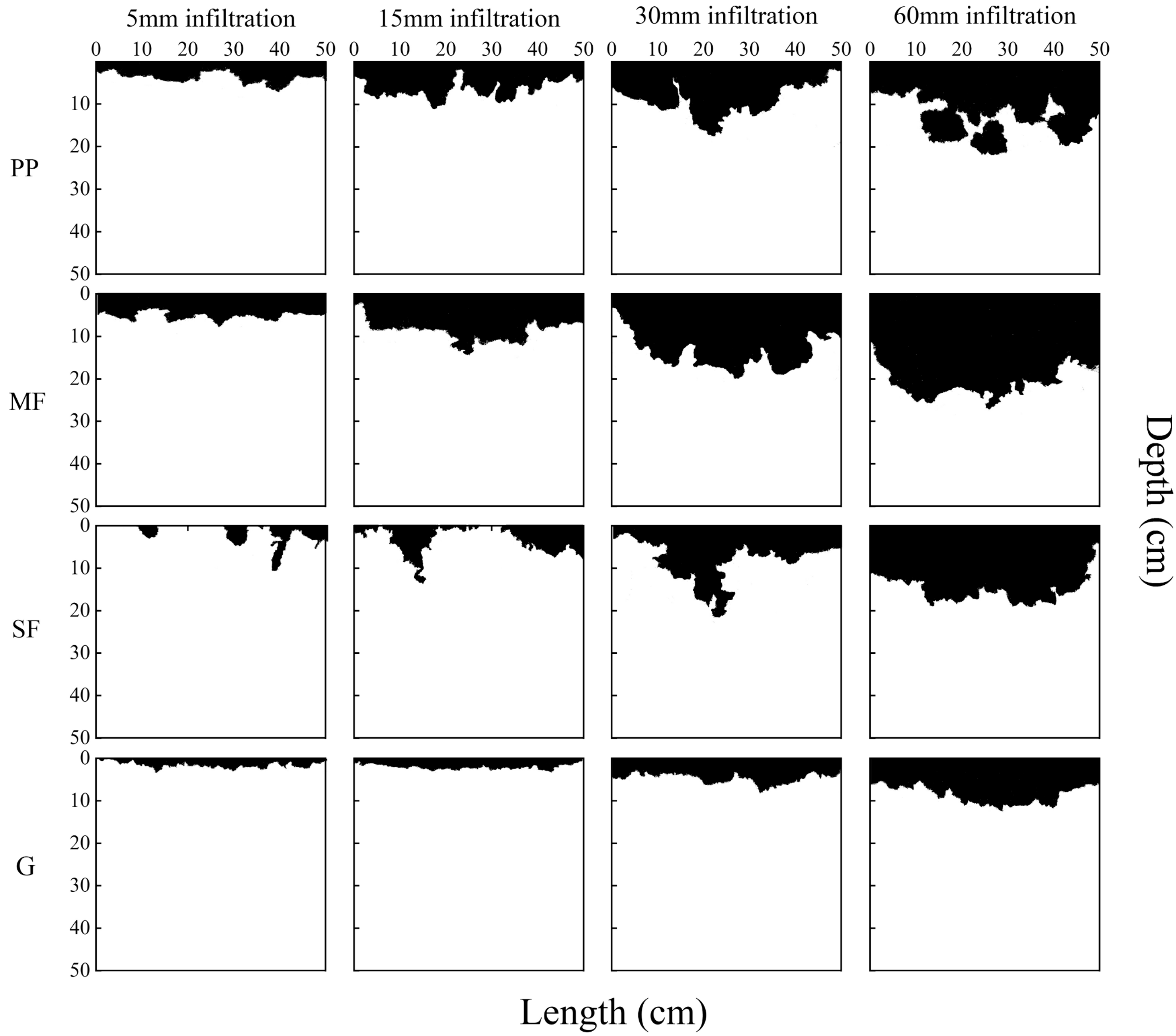

Fig. 7. Distribution of infiltrated water in soil profiles.

Spatial variability in preferential flow was analysed by calculating Li (Fig. 8). The overall Li increased with increasing infiltrated water under different vegetation covers. That is, the more water infiltrated into the soil, the greater the variability in preferential flow. In most cases, the variability of preferential flow was lower in the grassland than in the three woodlands. When the infiltration volume was 5 mm, the Li of SF was lower than that of PP and MF, and even lower than that of G. With the increase of infiltration water, its Li rose faster, and the Li of SF was 9.40 % higher than that of plantation forests under 60 mm infiltration water, which was 1.65 times higher than that of itself under 5 mm infiltration water. The Li of MF was higher than that of PP as a whole, and the difference between the two was greatest when the infiltration volume was 15 mm, and the Li difference between the two was very small when the infiltration volume was 30 mm and 60 mm. In addition, the Li of MF was basically the same at 15, 30, and 60 mm of infiltration volume, while the Li of PP was basically the same at 30 and 60 mm of infiltration volume.

3.3. *Influence factors of infiltration*

Vegetation restoration acts on surrounding environmental factors to alter infiltration characteristics and infiltration patterns. Matrix flow and preferential flow together constituted the total water infiltration in the soil, and they interacted with each other (Fig. 9). DC, as the most intuitive indicator of the overall water infiltration, interacted with other indicators of matrix flow and preferential flow. Under the typical vegetation in the Loess Plateau, DC was positively correlated with U_F, and the correlation coefficients were higher than 0.8. DC was negatively correlated with P_F, which means that the preferential flow phenomenon was not obvious when the water contact with the soil was more complete. In addition, P_F was not significantly correlated with Li, that is, there was no absolute correlation between the degree of development of preferential flow and its variability.

Environmental factors are closely related to infiltration and the formation of preferential flow. The amount of infiltration water significantly promoted the overall extent and depth of infiltration, as well as the depth of matrix flow and the variability degree of preferential flow,

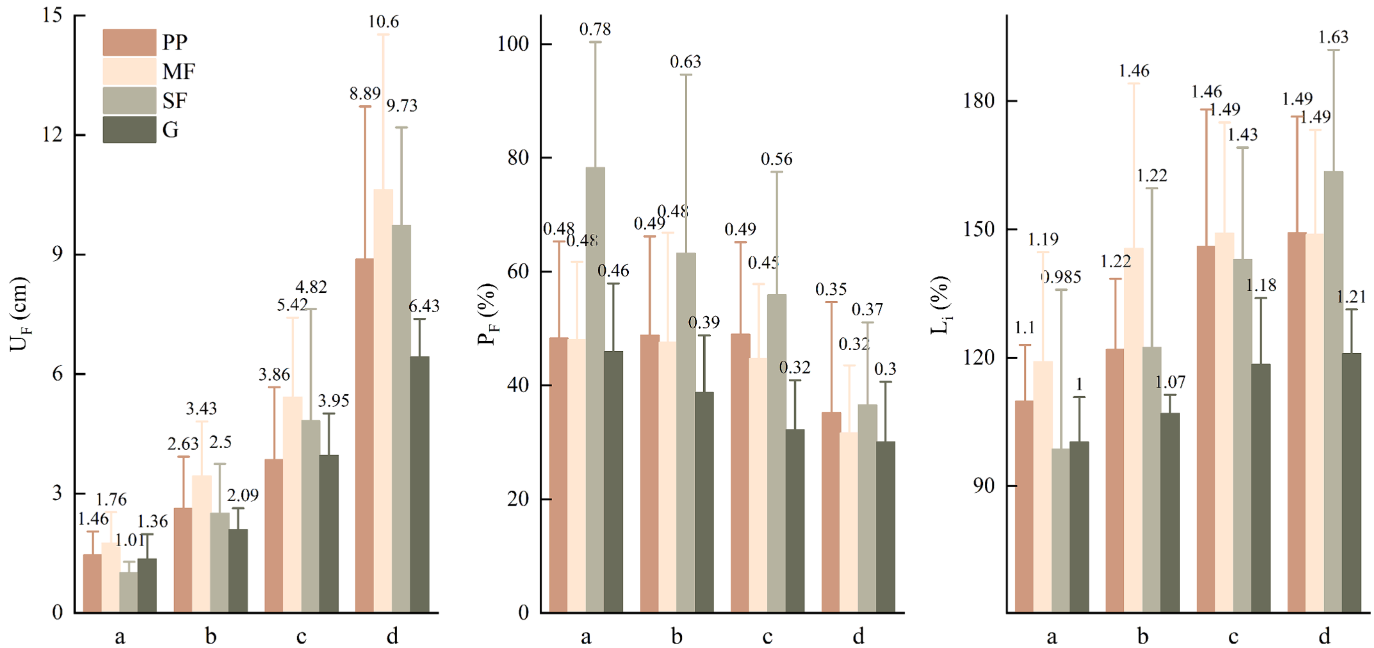

Fig. 8. Characteristics of matrix flow and preferential flow. a, b, c, d represent infiltration volumes of 5 mm, 15 mm, 30 mm and 60 mm, respectively.

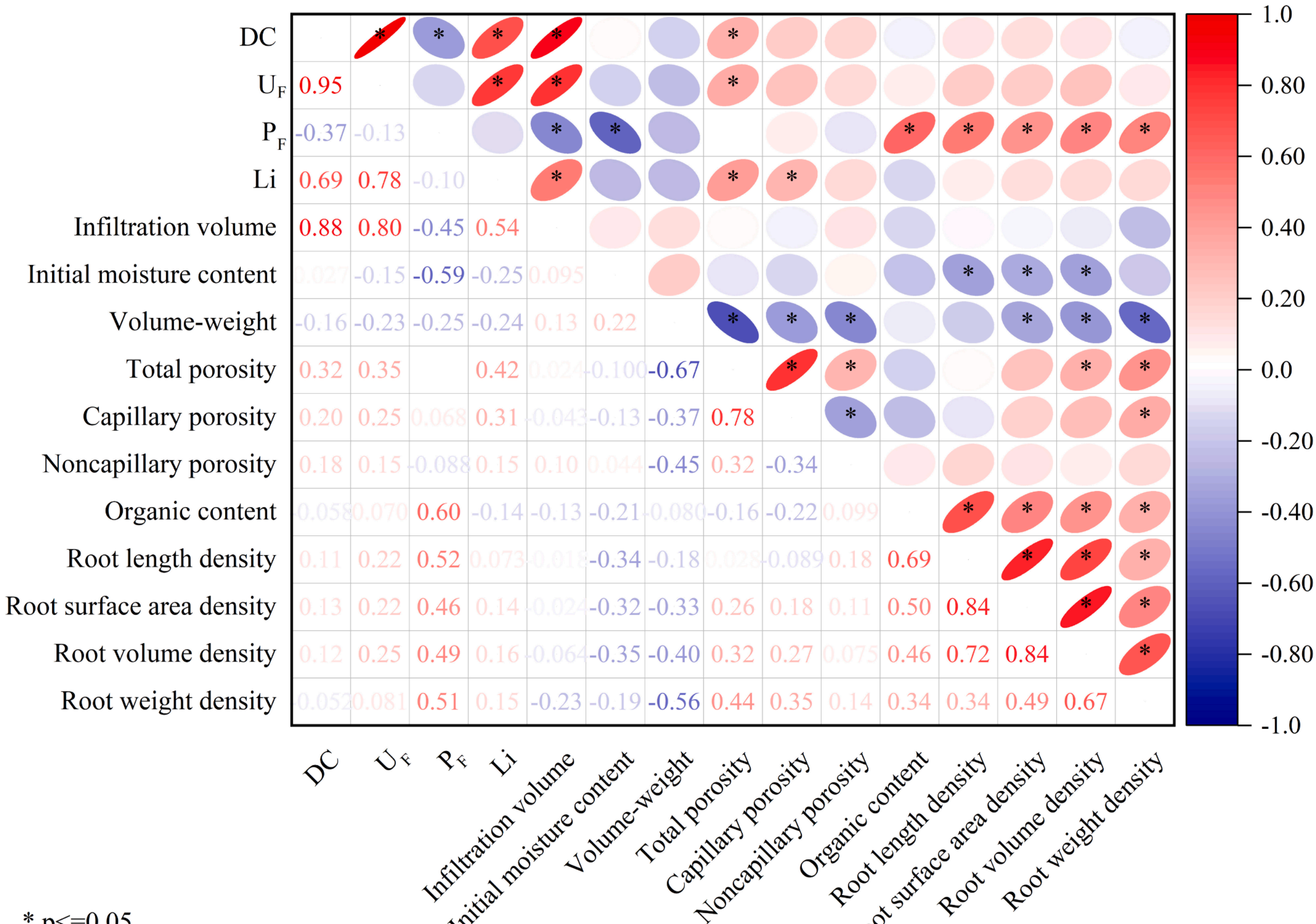

Fig. 9. Influence factors of infiltration. "*" indicates a significant correlation ($p < 0.05$), where red indicates a positive correlation and blue a negative correlation. The darker colour and smaller shape of the ellipse indicate a stronger correlation.

but inhibited the degree of preferential flow development. The initial water content of the soil significantly influenced P_F, with a correlation of −0.59.

The physical structure of the soil mainly affected DC, U_F and Li. The total porosity of the soil was significantly and positively correlated with DC, U_F and Li. The capillary porosity of the soil was significantly and positively correlated with Li, while it had no significant effect on the extent and depth of water infiltration. The non-capillary porosity of the soil was not significantly associated with water infiltration in this study. In addition, organic matter, as an important indicator of soil chemical properties, showed a significant positive correlation with P_F but had no significant effect on other infiltration indicators.

The large number of root systems distributed in the soil also influences water infiltration. However, in this study, the root system of plants only had remarkable influence on the development degree of preferential flow. Root length density, root surface area density, root volume density, and root weight density showed significant positive correlations with P_F, while there was no significant correlation with other infiltration indicators.

4. Discussion

4.1. Effects of vegetation restoration on infiltration characteristics in China's Loess Plateau

The large-scale and extensive afforestation program in the Loess Plateau of China has been implemented mainly to mitigate severe soil erosion and stabilize the local ecosystem (Cui et al., 2023). After long-term (more than 20 years) vegetation restoration, dramatic changes in soil properties have been found, and the infiltration capacity of soil and the movement of water in soil after precipitation have also changed significantly with the land use change and land coverage (Liao et al., 2023).

In the study, through the qualitative observation of stained profiles after simulated rainfall infiltration and the quantitative calculation of related parameters, it can be found that the restoration of forest vegetation can promote the infiltration of rainfall in general, which is manifested by the fact that under the same amount of infiltrated water (1) the area of contact between the water and the soil in the forests was larger than that in the grassland, and (2) the depth of the infiltration in the forests was larger than that in the grassland. This means that in the study area, the existing afforestation can significantly promote infiltration, and when rainfall occurs, water can fully infiltrate the soil and recharge the deeper soil layers. However, even under high precipitation, the depth of water recharge was still limited to surface soil (upper 30 cm), with limited recharge of deeper soil moisture, which may lead to the formation of dry soil layers at deeper soil scales (Feng et al., 2023).

Different attributes of vegetation restoration types lead to differences in soil physicochemical properties, soil hydrological properties, and root distribution in the soil, so although various vegetation restoration types can improve soil infiltration properties, their infiltration characteristics are not consistent (Wang et al., 2022a). *Robinia pseudoacacia* and *Pinus tabulaeformis* are commonly used in afforestation projects in the Loess Plateau, and for the two plantation forests in this study, the mixed forest of *Robinia pseudoacacia* and *Pinus tabulaeformis* had a better ability to infiltrate and absorb water than the pure forest of *Pinus tabulaeformis*. Through attribution analysis shown that this was due to the higher total porosity of the soil in the mixed forest. *Robinia pseudoacacia* has a strong primary root system, and the lateral roots of *Pinus tabulaeformis* are widely distributed in the 0 ~ 20 cm soil layer (Mei et al., 2022), and the forces generated by the stronger primary root system during the growth process will make the originally dense soil loose and porous, and promote the movement of water in the soil (Colombi et al., 2021).

4.2. Effects of vegetation restoration on infiltration patterns in China's Loess Plateau

Afforestation not only improves the infiltration properties of soil but also changes infiltration patterns. After long-term vegetation restoration in the Loess Plateau, forest vegetation cover significantly increases the diversity of species in the understory, and under the multiple effects of plant roots, burrowing animals, etc., more and more complex three-dimensional pore structures are formed in the soil, constituting preferential paths in the soil and contributing to the formation of preferential flow. (Maier et al., 2019).

In this study, forested soil shows a more pronounced preferential flow phenomenon compared to grassland, confirming the conjecture that vegetation restoration is affecting infiltration patterns. However, the development of preferential flow was not the same due to differences in soil infiltration properties. The degree of preferential flow development in the natural secondary forest was significantly higher than that in the plantation forest, especially when the amount of infiltration water was low. When there are more preferential paths in the soil, their own larger infiltration pore channels and the alteration of the surrounding soil structure during their formation will cause water to pool more towards these preferential paths, and if the infiltration water amount is low, the water cannot be recharged to the matrix domain at the same time (Kan et al., 2019), which will lead to insufficient uptake of water by the top soil layer. Therefore, the preferential flow ratio tends to decrease with increasing infiltration water. In contrast, for planted forests with relatively few preferential paths, the response of preferential flow ratios to infiltration water was not significant.

An attribution analysis yielded that, in addition to infiltration water, preferential flow ratio was simultaneously influenced by soil moisture content, organic matter content, and root system. Compared to grassland, lower soil moisture content in woodlands leads to soil shrinkage and the formation of wide and deep cracks in the soil (Ma et al., 2015), which become preferential paths for water infiltration.

In the loess region with relatively homogeneous and deep soil, the plant root system distributed in soil is the main source of preferential paths. Especially when long and thick roots are more abundant, they form a connected pore structure in soil (Wu et al., 2021), which is conducive to the development of preferential flow. In natural secondary forests, due to lower canopy density and sufficient light within the forest, more opportunities are provided for the growth and development of understory plants (Fu et al., 2017). In addition, the study found the natural secondary forest had more root distributions in the shallow soil layer than planted forests, and these roots enhance the connectivity of the preferential paths with the ground surface. When water infiltrates into the soil, the water supply of the soil at the leading edge of the wet front is recharged by the drainage of the upper wet soil layer, which means that infiltration in the upper layer influences infiltration of the lower layer to a certain extent. For planted forests with relatively weak connectivity of the preferential paths to the surface, water cannot preferentially infiltrate as it flows from the upper soil matrix domain to the lower macroporous domain due to reduced water potential gradient pressure, making it difficult for preferential flow to form (Nobles et al.,2010).

Soil organic matter content also promoted the development of preferential flow in this study. Soil organic matter interacts with vegetation, leaf litter and root decay secretions are important sources of organic matter. Organic matter also promotes vegetation and root growth (Guidi et al., 2023), thus influencing the formation and development of preferential flow.

In the study, although the degree of development of preferential flow was always higher in natural secondary forest than in planted forests, in most cases, preferential flow in natural secondary forest had lower spatial variability. This may also be due to the fact that a well-connected network structure of preferential paths has been formed in secondary forest, and when water flows move through them, there is a stronger interaction between water flows and macroporous domains, resulting in more concentrated and continuous infiltration paths (Li et al., 2019) and lower spatial variability.

4.3. Implications for vegetation restoration and further infiltration research in the Loess Plateau region

In the ecologically fragile Loess Plateau region, it is important to improve the quality of forests and maximize their seepage-promoting and water-holding functions (Wang et al., 2022b). At the initial stage of afforestation on the Loess Plateau, a single dominant tree species afforestation model is often used, which has significant advantages in terms of initial growth and early stand benefits; However, the ecological problems associated with the use of a single dominant tree species in plantation forests have gradually emerged, including reduced resource utilization efficiency, increased competition between dominant species and understory plants, and excessive soil moisture depletion (Shen et al., 2020). Based on the results of this study, it can also be found that under a pure plantation forest, the ability of soil water absorption was weaker than that of mixed plantation forest and natural secondary forest which were not a single tree species. To cope with this challenge, mixed forests can be cultivated to create a complex and stable forest community and enhance the infiltration and water retention function of forests.

In the study, the water under the mixed forest of *Pinus tabulaeformis* and *Robinia pseudoacacia* was able to infiltrate the soil more fully and recharge the deep soil layer, and when the precipitation was low, there was no phenomenon that the water infiltrated a lot along the preferential paths and failed to fully recharge the surface soil layer. Therefore, the *Pinus tabulaeformis* – *Robinia pseudoacacia* hybrid model may be more suitable for the current situation in the study area in terms of water conservation. Thus, this study suggests that the reliance on a single dominant tree species should be gradually reduced in afforestation projects on the Loess Plateau to ensure a more rational and effective utilization of natural resources.

This study mainly used the bright blue staining tracer experiment combined with the indoor analysis experiment, which has the advantages of convenient operation, no pollution, easy to identify, etc., and can reflect the infiltration pattern and characteristics through the distribution pattern of water in the soil profile very intuitively. However, this method also has limitations, the most important of which is that it cannot reflect the whole movement process of water infiltration into the soil, making the infiltration experiment still a 'black box' experiment. Therefore, in further infiltration studies, more consideration should be given to the dynamic processes of matrix infiltration and preferential flow infiltration. In addition, more microscopic factors, such as the size, number, connectivity and curvature of water infiltration pathways, should also be considered later when analysing whether and how vegetation restoration affects infiltration. These subsequent studies will provide a more comprehensive and penetrating understanding of infiltration and overall hydrological processes in the Loess Plateau.

5. Conclusion

The effect of vegetation restoration on infiltration was explored by investigating the infiltration characteristics and infiltration patterns of typical vegetation restoration types on the Loess Plateau. The results showed that vegetation restoration could increase soil porosity, organic matter content and root distribution, and improved the ranges of movement of water in soil after precipitation infiltration so that the external water source could be recharged to the deeper soil layers. The forest vegetation cover also led to preferential flow phenomenon of infiltration processes. The preferential flow in the natural secondary forest was more obvious than that in the plantation forest, especially in the case of small precipitation. The contact between soil and water was more complete in the mixed plantation forest of *Pinus tabulaeformis* and *Robinia pseudoacacia*, and the effect of soil infiltration and water absorption was better than that of the pure *Pinus tabulaeformis* plantation forest. From the aspect of promoting infiltration and water conservation, this model of *Pinus tabulaeformis* – *Robinia pseudoacacia* hybrid was more suitable for the current situation in the study area.

CRediT authorship contribution statement

Ning Guan: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Huaxing Bi:** Supervision, Conceptualization. **Yilin Song:** Investigation. **Shanhong Lu:** Investigation. **Dandan Lin:** Investigation. **Jindan Han:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

Beven, K., Germann, P., 1982. Macropores and water flow in soil. Water Resour. Res. 18 (5), 1311–1325. https://doi.org/10.1029/WR018i005p01311.

Chen, M.T., Yang, X.Q., Zhang, X.T., Bai, Y., Shao, M.A., Wei, X.R., Jia, Y.H., Wang, Y.Q., Jia, X.X., Zhu, Y.J., Zhang, Q.Y., Zhu, X.C., Li, T.C., 2024. Response of soil water to long-term revegetation, topography, and precipitation on the Chinese Loess Plateau. Catena 236, 107711. https://doi.org/10.1016/j.catena.2023.107711.

Cheng, Z., Yu, B.F., Fu, S.H., Liu, G., 2018. Comparative evaluation of three infiltration models for runoff and erosion prediction in the Loess Plateau region of China. Hydrol. Process. 49 (5), 1484–1497. https://doi.org/10.2166/nh.2017.003.

Colombi, T., Kirchgessner, N., Vuilleumier, C., Iseskog, D., Alexandersson, S., Larsbo, M., Keller, T., 2021. A time-lapse imaging platform for quantification of soil crack development due to simulated root water uptake. Soil Tillage Res. 205, 104769 https://doi.org/10.1016/j.still.2020.104769.

Cui Y.S., Pan, C.Z., 2023. Hydrological responses to litter density on runoff-infiltration patterns and water conservation in Pinus tabuliformis plantation. Journal of Hydrology. 619, 129293. https://doi.org/ 10.1016/J.JHYDROL.2023.129293.

Cui, Z., Wu, G.L., Huang, Z., Liu, Y., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. J. Hydrol. 578, 124023 https://doi.org/10.1016/j.jhydrol.2019.124023.

Feng, T.J., Wei, T.X., Keesstra, S.D., Zhang, J.J., Bi, H.X., Wang, R.S., Wang, P., 2023. Long-term effects of vegetation restoration on hydrological regulation functions and the implications to afforestation on the Loess Plateau. Agricultural and Forest Meteorology 330, 109313. https://doi.org/10.1016/j.agrformet.2023.109313.

Fu, B.J., Wang, S., Liu, Y., Liu, J.B., Liang, W., Miao, C.Y., 2017. Hydrogeomorphic ecosystem responses to natural and anthropogenic changes in the loess plateau of China. Annu. Rev. Earth Planet. Sci. 45 (1), 223–243. https://doi.org/10.1146/annurev-earth-063016-020552.

Guan, N., Cheng, J.H., Shi, X.Q., 2023. Preferential flow and preferential path characteristics of the typical forests in the karst region of southwest China. Forests 14 (6), 1248. https://doi.org/10.3390/f14061248.

Guidi, C., Lehmann, M.M., Meusburger, K., Saurer, M., Vitali, V., Peter, M., Brunner, I., Hagedorn, F., 2023. Tracing sources and turnover of soil organic matter in a long-term irrigated dry forest using a novel hydrogen isotope approach. Soil Biol. Biochem. 183, 109113 https://doi.org/10.1016/j.soilbio.2023.109113.

Guo, L., Liu, Y., Wu, G.L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.Q., Tian, F.P., He, H.H., 2019. Preferential water flow: influence of alfalfa (Medicago sativa L.) decayed root channels on soil water infiltration. J. Hydrol. 578, 124019 https://doi.org/10.1016/j.jhydrol.2019.124019.

Hao, M.Z., Zhang, J.C., Meng, M.J., Chen, H.Y.H., Guo, X.P., Liu, S.L., Ye, L.X., 2019. Impacts of changes in vegetation on saturated hydraulic conductivity of soil in subtropical forests. Sci Rep. 9, 8372. https://doi.org/10.1038/s41598-019-44921-w.

Jia, S., Zhu, W., 2020. China's achievements of water governance over the past seven decades. Int. J. Water Resour. Dev. 36 (2–3), 292–310. https://doi.org/10.1080/07900627.2019.1709422.

Jiang, X.J., Liu, W., Wu, J., Wang, P., Liu, C., Yuan, Z.Q., 2017. Land degradation controlled and mitigated by rubber-based agroforestry systems through optimizing soil Physical conditions and water supply mechanisms: a case study in Xishuangbanna. China. Land Degradation & Development. 28 (7), 2277–2289. https://doi.org/10.1002/ldr.2757.

Kan, X.Q., Cheng, J.H., Hu, X.J., Zhu, F.F., Li, M., 2019. Effects of grass and forests and the infiltration amount on preferential flow in karst regions of China. Water 11 (8), 1634. https://doi.org/10.3390/w11081634\.

Li, X., Lu, Y.D., Zhang, X.Z., Fan, W., Lu, Y.C., Pan, W.S., 2019. Quantification of macropores of Malan loess and the hydraulic significance on slope stability by X-ray computed tomography. Environ. Earth Sci. 78 (16), 1–19. https://doi.org/10.1007/s12665-019-8527-2.

Li, F.D., Song, X.F., Tang, C.Y., 2007. Tracing infiltration and recharge using stable isotope in Taihang Mt. North China. Environmental Geology. 53, 687–696. https://doi.org/10.1007/s00254-007-0683-0.

Liao, Y., Dong, L.B., Li, A., Lv, W.W., Wu, J.Z., Zhang, H.L., Bai, R.H., Liu, Y.L., Li, J.W., Shangguan, Z.P., Deng, L., 2023. Soil physicochemical properties and crusts regulate the soil infiltration capacity after land-use conversions from farmlands in semiarid areas. J. Hydrol. 626 (PA), 130283 https://doi.org/10.1016/j.jhydrol.2023.130283.

Liu, Y., Zhang, Y.H., Xie, L.M., Zhao, S.Q., Dai, L.Y., Zhang, Z.M., 2021. Effect of soil characteristics on preferential flow of Phragmites australis community in Yellow River delta. Ecol. Ind. 125, 107486 https://doi.org/10.1016/j.ecolind.2021.107486.

Luo, Z.T., Niu, J.Z., Zhang, L., Chen, X.W., Zhang, W., Xie, B.Y., Du, J., Zhu, Z.J., Wu, S. S., Li, X., 2019. Roots-Enhanced preferential flows in deciduous and coniferous forest soils revealed by dual-tracer experiments. J. Environ. Qual. 48 (1), 136–146. https://doi.org/10.2134/jeq2018.03.0091.

Ma, R.M., Cai, C.F., Li, Z.X., Wang, J.G., Xiao, T.Q., Peng, G.Y., 2015. Evaluation of soil aggregate microstructure and stability under wetting and drying cycles in two Ultisols using synchrotron-based X-ray micro-computed tomography. Soil Tillage Res. 149, 1–11. https://doi.org/10.1016/j.still.2014.12.016.

Maier, F., van Meerveld, I., Greinwald, K., Gebauer, T., Lustenberger, F., Hartmann, A., Musso, A., 2019. Effects of soil and vegetation development on surface hydrological properties of moraines in the Swiss Alps. Catena 187 (C), 104353. https://doi.org/10.5194/egusphere-egu21-830.

Mei, X.M., Ma, L., 2022. Effect of afforestation on soil water dynamics and water uptake under different rainfall types on the Loess hillslope. Catena 213, 106216. https://doi.org/10.1016/j.catena.2022.106216.

Mei, X.M., Zhu, Q.K., Ma, L., Zhang, D., Wang, Y., Hao, W.J., 2018. Effect of stand origin and slope position on infiltration pattern and preferential flow on a Loess hillslope. Land Degrad. Dev. 29 (5), 1353–1365. https://doi.org/10.1002/ldr.2928.

Nobles, M.M., Wilding, L.P., Lin, H.S., 2010. Flow pathways of bromide and Bright Blue FCF tracers in caliche soils. J. Hydrol. 393 (1), 114–122. https://doi.org/10.1016/j.jhydrol.2010.03.014.

Ren, Z.P., Zhu, L.J., Wang, B., Cheng, S.D., 2016. Soil hydraulic conductivity as affected by vegetation restoration age on the Loess Plateau. China. Journal of Arid Land. 8 (4), 546–555. https://doi.org/10.1007/s40333-016-0010-2.

Shen, M.S., Zhang, J.J., Zhang, S.H., Zhang, H.B., Sun, R.X., Zhang, Y.Z., 2020. Seasonal variations in the influence of vegetation cover on soil water on the loess hillslope. J. Mt. Sci. 17 (9), 2148–2160. https://doi.org/10.1007/s11629-019-5942-5.

Wang, N., Bi, H.X., Cui, Y.H., Zhao, D.Y., Hou, G.R., Yun, H.Y., Liu, Z.H., Lan, D.Y., Jin, C.A., 2022b. Optimization of stand structure in Robinia pseudoacacia Linn. based on soil and water conservation improvement function. Ecol. Ind. 136, 108671 https://doi.org/10.1016/j.ecolind.2022.108671.

Wang, N., Bi, H.X., Peng, R.D., Zhao, D.Y., Liu, Z.H., 2023. Disparities in soil and water conservation functions among different forest types and implications for afforestation on the Loess Plateau. Ecol. Ind. 155 (5), 110935 https://doi.org/10.1016/j.ecolind.2023.110935.

Wang, C., Tang, C.J., Fu, B.J., Lü, Y.H., Xiao, S.S., Zhang, J., 2022a. Determining critical thresholds of ecological restoration based on ecosystem service index: a case study in the Pingjiang catchment in southern China. J. Environ. Manage. 303, 114220 https://doi.org/10.1016/j.jenvman.2021.114220.

Wu, G.L., López Vicente, M., Huang, Z., Cui, Z., Liu, Y., 2020. Preferential water flow through decayed root channels enhances soil water infiltration: evaluation in distinct vegetation types under semi-arid conditions. Hydrol. Earth Syst. Sci. 1–22 https://doi.org/10.5194/hess-2020-266.

Wu, G.L., Cui, Z., Huang, Z., 2021. Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. Geoderma 404, 115289. https://doi.org/10.1016/j.geoderma.2021.115289.

Yao, J.J., Cheng, J.H., Sun, L., Zhang, X., Zhang, H.J., 2017. Effect of Antecedent Soil Water on Preferential Flow in Four Soybean Plots in Southwestern China. Soil Sci. 2017 https://doi.org/10.1097/SS.0000000000000198.

Zhang, Y.Y., Wu, P.T., Zhao, X.N., Li, P., 2012. Evaluation and modelling of furrow infiltration for uncropped ridge–furrow tillage in Loess Plateau soils. Soil Research, 50(5), 360-370. https://doi.org/ 10.1071/SR12061.

Zhang, J., Lei, T.W., Qu, L.Q., Chen, P., Gao, X.F., Chen, C., Yuan, L.L., Zhang, M.L., Su, G.G., 2017. Method to measure soil matrix infiltration in forest soil. J. Hydrol. 552, 241–248. https://doi.org/10.1016/j.jhydrol.2017.06.032.

Zhang, J.Y., Liu, T.X., Duan, L.M., Chen, Z.X., Wang, Y.X., Li, Y.K., Zhao, X.Y., Wang, G. Q., Singh, V.P., 2022. Processes of preferential flow in a eurasian steppe under different scenarios. J. Hydrol. 612, 128166 https://doi.org/10.1016/j.jhydrol.2022.128166.

Zhang, Y.Y., Zhao, W.Z., Li, X.B., Jia, A., Kang, W.R., 2021. Contribution of soil macropores to water infiltration across different land use types in a desert–oasis ecoregion. Land Degrad. Dev. 32, 1751–1760. https://doi.org/10.1002/ldr.3823.

Zhao, M.Q., Li, D.H., Huang, Y., Deng, Y.S., Yang, G., Lei, T.W., Huang, Y.H., 2022. Soil matrix infiltration characteristics in differently aged eucalyptus plantations in a southern subtropical area in China. Catena 217, 106490. https://doi.org/10.1016/j.catena.2022.106490.